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global_data = {

    # Attribute Names (unchanged)
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "conductor_gauge_mm2",
    "attribute_4": "cable_length_m",
    "attribute_5": "service_amperage_A",

    # Attribute Directions (unchanged)
    "direction_attribute_1": "negative",
    "direction_attribute_2": "positive",
    "direction_attribute_3": "positive",
    "direction_attribute_4": "negative",
    "direction_attribute_5": "positive",

    # Global Attribute Bounds (UPDATED for universality)

    # Attribute 1: total_cost (USD)
    # Old range: $16.00 – $107.30 (too tight — only reflects current 8 products)
    # Lower bound: smallest conceivable drip loop — minimal gauge Al cable, near-zero labor
    #           (e.g., rural sub-Saharan Africa or informal economy install)
    # Upper bound: largest gauge Cu cable (e.g., 350 kcmil) + premium labor market (e.g.,
    #           Scandinavia or Switzerland) + long drip loop + premium fittings
    "attribute_1_min": 5.00,           # USD — absolute floor for any viable drip loop product
    "attribute_1_max": 300.00,        # USD — ceiling for premium Cu, large gauge, h

    # Attribute 2: material_continuity (binary — no change, already universal)
    # 0 = aluminum history | 1 = copper history; these are the only two possible values
    "attribute_2_min": 0,
    "attribute_2_max": 1,

    # Attribute 3: conductor_gauge_mm²
    # Old range: 16.0 – 107.2mm² (current dataset only)
    # Lower bound: 4mm² = smallest code-compliant SEC conductor (IS 694 / IEC fo
    #           feasible in very low-load rural electrification programs)
    # Upper bound: 300mm² = large commercial/industrial service entrance (~600A class),
    #           well within IEC 60228 Class 2 stranded conductor range
    "attribute_3_min": 4.0,           # mm² — smallest practical SEC conductor
    "attribute_3_max": 300.0,        # mm² — large commercial service, still a drip loop config

    # Attribute 4: cable_length_m
    # Old range: 0.5 – 3.0m (practical current installs only)
    # Lower bound: 0.3m = bare minimum to physically form any downward loop at

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#           below this, water cannot reliably drip clear of the service head
# Upper bound: 6.0m = unusually tall service mast or long overhead service drop clearance
#           requirement over a steep slope or special structure (still a drip loop context)
"attribute_4_min": 0.30,      # m — physically minimum functional drip formation
"attribute_4_max": 6.00,      # m — maximum plausible for tall mast / special clearance

# Attribute 5: service_amperage_A
# Old range: 63 – 200A (current dataset only)
# Lower bound: 30A = smallest residential service found globally (e.g., single-room dwellings
#           in rural India, parts of Africa/SE Asia; IEC permits 230V/30A drops)
# Upper bound: 400A = maximum standard residential/light commercial service p
#           beyond this, drip loop SEC cables give way to switchgear/bus duct configurations
"attribute_5_min": 30,        # A — minimum viable residential service amperage globally
"attribute_5_max": 400,       # A — NEC/IEC residential ceiling before switchgear territory

# Global Material Prices (unchanged)
"copper_price_per_ton": 9500,
"aluminum_price_per_ton": 2500,
"copper_price_per_kg": 9.5,
"aluminum_price_per_kg": 2.5
}

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regional_data = [

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# United States
# NEC-governed, strong copper legacy from 1970s Al wiring failure stigma,
# 200A becoming universal standard for new construction and upgrades.
# Copper-dominant replacement market; cost matters but safety/continuity matters more.
{
  "region": "United States",
  "copper_product_market_share": 0.60, # Copper preferred in residential SE cable
  "aluminum_product_market_share": 0.40, # Al used but with termination care required
  "weight_attribute_1": 0.18,      # Cost: moderate — copper premium accepted
  "weight_attribute_2": 0.25,      # Material continuity: HIGH — legacy Cu stigma
  "weight_attribute_3": 0.25,      # Conductor gauge: HIGH — NEC 310.12 strict sizing
  "weight_attribute_4": 0.12,      # Cable length: LOW — short, constrained drip loop
  "weight_attribute_5": 0.20,      # Service amperage: HIGH — 200A standard dr
},                                # Sum: 0.18+0.25+0.25+0.12+0.20 = 1.00 ✓

# China
# IEC/GB standards, predominantly new construction (low material history effect),
# strong cost sensitivity at scale, aluminum increasingly adopted in new builds.
{
  "region": "China",

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    "copper_product_market_share": 0.40, # Cu still used in premium/urban builds
    "aluminum_product_market_share": 0.60, # Al dominant in cost-optimized construction
    "weight_attribute_1": 0.30,          # Cost: HIGH — large-scale cost optimization
    "weight_attribute_2": 0.10,          # Material continuity: LOW — new construction d
    "weight_attribute_3": 0.25,          # Conductor gauge: HIGH — GB/IEC comp
    "weight_attribute_4": 0.15,          # Cable length: MODERATE — varied buildin
    "weight_attribute_5": 0.20,          # Service amperage: MODERATE — 100A standard
  },                                     # Sum: 0.30+0.10+0.25+0.15+0.20 = 1.00 ✓

# India
# Most cost-sensitive market, mixed new/retrofit base, IS 694/IEC-aligned standards,
# aluminum dominant due to copper price premium, lower amperage requirements.
{
  "region": "India",
  "copper_product_market_share": 0.25, # Cu used in premium urban/industrial builds
  "aluminum_product_market_share": 0.75, # Al strongly dominant due to cost pressure
  "weight_attribute_1": 0.35,          # Cost: HIGHEST — primary decision driver
  "weight_attribute_2": 0.15,          # Material continuity: LOW-MOD — mixed installed base
  "weight_attribute_3": 0.20,          # Conductor gauge: MODERATE — IS 694 compliance
  "weight_attribute_4": 0.15,          # Cable length: MODERATE — variable building stock
  "weight_attribute_5": 0.15,          # Service amperage: LOWEST — narrow 63A range
},                                     # Sum: 0.35+0.15+0.20+0.15+0.15 = 1.00 ✓

# Europe
# IEC/HD 60364 culture of rigorous compliance, copper traditional in residential,
# highest labor costs, conductor sizing precision weighted most heavily by engineers.
{
  "region": "Europe",
  "copper_product_market_share": 0.55, # Cu slightly dominant in residential SEC
  "aluminum_product_market_share": 0.45, # Al accepted, growing in new builds
  "weight_attribute_1": 0.15,          # Cost: LOWEST — quality/compliance leads
  "weight_attribute_2": 0.20,          # Material continuity: MOD — mature infrastructure
  "weight_attribute_3": 0.30,          # Conductor gauge: HIGHEST — IEC spec culture
  "weight_attribute_4": 0.15,          # Cable length: MODERATE — stand
  "weight_attribute_5": 0.20,          # Service amperage: MOD — 100A st
},                                     # Sum: 0.15+0.20+0.30+0.15+0.20 = 1.00 ✓
]

product_data = [

# =====
# UNITED STATES
# Standard: NEC Article 230 / 310.12 | Voltage: 600V | Service: 120/240V split-phase
# Cu: 2/0 AWG (67.4mm²) | Al: 4/0 AWG (107.2mm²) — both rated 200A per N

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Conductor mass: Cu = $3 \times 67.4\text{mm}^2 + 1 \times 53.5\text{mm}^2 = 255.7\text{mm}^2$ total | Al = 3×107

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# =====  
  
{  
  "region": "United States",  
  "dominant material": "cu",  
  "product": "Copper SER Service Entrance Cable, 2/0 AWG, 3C+G, 600V, 1.5m Drip Loop",  
  "copper_kg": 3.40,      #  $255.7\text{mm}^2 \times 1.5\text{m} \times 8.960\text{ g/cm}^3 = 3,437\text{g} \rightarrow 3.40\text{ kg}$   
  "aluminum_kg": 0.00,  
  "total_mass_kg": 5.50,  #  $3.40\text{kg Cu} \div 0.62$  (Cu content fraction of SER cable)  
  "non_material_cost_per_unit": 75.00, # USD: US labor + PVC jacket + weather  
  "attribute_2_value": 1,   # material_continuity: copper history  $\rightarrow 1$   
  "attribute_3_value": 67.4, # conductor_gauge_mm2: 2/0 AWG =  $67.43\text{mm}^2$   
  "attribute_4_value": 1.50, # cable_length_m: standard drip loop  
  "attribute_5_value": 200,  # service_amperage_A: US 200A standard  
},  
{  
  "region": "United States",  
  "dominant material": "al",  
  "product": "Aluminum SER Service Entrance Cable, 4/0 AWG, 3C+G, 600V, 1.5m Drip Loop",  
  "copper_kg": 0.00,  
  "aluminum_kg": 1.60,      #  $389\text{mm}^2 \times 1.5\text{m} \times 2.700\text{ g/cm}^3 = 1,576\text{g} \rightarrow 1.60\text{ kg}$   
  "total_mass_kg": 3.10,    #  $1.60\text{kg Al} \div 0.52$  (Al content fraction of SER cable)  
  "non_material_cost_per_unit": 75.00, # Same US labor; Al termination compound included  
  "attribute_2_value": 0,    # material_continuity: aluminum history  $\rightarrow 0$   
  "attribute_3_value": 107.2, # conductor_gauge_mm2: 4/0 AWG =  $107.2\text{mm}^2$   
  "attribute_4_value": 1.50, # cable_length_m  
  "attribute_5_value": 200,  # service_amperage_A  
},
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# =====  
# CHINA  
# Standard: GB/T 9330 / IEC 60228 | Voltage: 0.6/1kV | Service: 220V single-phase, 100A  
# Cu:  $35\text{mm}^2$  phase conductor | Al:  $50\text{mm}^2$  phase conductor (IEC equivalency for 100A)  
# Conductor mass: Cu =  $3 \times 35\text{mm}^2 + 1 \times 25\text{mm}^2 = 130\text{mm}^2$  total | Al =  $3 \times$   
# =====
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{  
  "region": "China",  
  "dominant material": "cu",  
  "product": "Copper Service Entrance Cable,  $35\text{mm}^2$ , 3C+G, 0.6/1kV, 1.5m Drip Loop",  
  "copper_kg": 1.75,      #  $130\text{mm}^2 \times 1.5\text{m} \times 8.960\text{ g/cm}^3 = 1,747\text{g} \rightarrow 1.75\text{ kg}$   
  "aluminum_kg": 0.00,  
  "total_mass_kg": 2.80,  #  $1.75\text{kg Cu} \div 0.62$ 
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"non_material_cost_per_unit": 25.00, # USD: China labor + XLPE/PVC insulation + fittings
"attribute_2_value": 1, # material_continuity: copper history → 1
"attribute_3_value": 35.0, # conductor_gauge_mm2: IEC standard 35mm²
"attribute_4_value": 1.50, # cable_length_m
"attribute_5_value": 100, # service_amperage_A: China standard residential 100A
},
{
  "region": "China",
  "dominant material": "al",
  "product": "Aluminum Service Entrance Cable, 50mm², 3C+G, 0.6/1kV, 1.5m Drip Loop",
  "copper_kg": 0.00,
  "aluminum_kg": 0.75, # 185mm² × 1.5m × 2.700 g/cm³ = 749g → 0.75 kg
  "total_mass_kg": 1.40, # 0.75kg Al ÷ 0.52
  "non_material_cost_per_unit": 25.00,
  "attribute_2_value": 0, # material_continuity: aluminum history → 0
  "attribute_3_value": 50.0, # conductor_gauge_mm2: IEC standard 50mm²
  "attribute_4_value": 1.50, # cable_length_m
  "attribute_5_value": 100, # service_amperage_A
},

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# =====
# INDIA
# Standard: IS 694 / IEC 60228 | Voltage: 1.1kV | Service: 230V single-phase, 63A
# Cu: 16mm² phase conductor | Al: 25mm² phase conductor (IS/IEC equivalency for 63A)
# Conductor mass: Cu = 3×16mm² + 1×10mm² = 58mm² total | Al = 3×25m
# =====

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{
  "region": "India",
  "dominant material": "cu",
  "product": "Copper Service Entrance Cable, 16mm², 3C+G, 1.1kV, 1.5m Drip Loop",
  "copper_kg": 0.80, # 58mm² × 1.5m × 8.960 g/cm³ = 780g → 0.80 kg
  "aluminum_kg": 0.00,
  "total_mass_kg": 1.30, # 0.80kg Cu ÷ 0.62
  "non_material_cost_per_unit": 15.00, # USD: India labor + PVC insulation + fittings
  "attribute_2_value": 1, # material_continuity: copper history → 1
  "attribute_3_value": 16.0, # conductor_gauge_mm2: IS 694 standard 16mm²
  "attribute_4_value": 1.50, # cable_length_m
  "attribute_5_value": 63, # service_amperage_A: India standard residential 63A
},
{
  "region": "India",
  "dominant material": "al",
  "product": "Aluminum Service Entrance Cable, 25mm², 3C+G, 1.1kV, 1.5m Drip Loop",

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    "copper_kg": 0.00,
    "aluminum_kg": 0.40,      #  $91\text{mm}^2 \times 1.5\text{m} \times 2.700\text{ g/cm}^3 = 369\text{g} \rightarrow 0.40\text{ kg}$ 
    "total_mass_kg": 0.80,    #  $0.40\text{kg Al} \div 0.52$ 
    "non_material_cost_per_unit": 15.00,
    "attribute_2_value": 0,    # material_continuity: aluminum history  $\rightarrow 0$ 
    "attribute_3_value": 25.0, # conductor_gauge_mm2: IS 694 standard  $25\text{mm}^2$  (Al e
    "attribute_4_value": 1.50, # cable_length_m
    "attribute_5_value": 63,   # service_amperage_A
  },

# =====
# EUROPE
# Standard: IEC 60228 / HD 60364 | Voltage: 0.6/1kV | Service: 230V single-phase, 100A
# Cu:  $35\text{mm}^2$  phase conductor | Al:  $50\text{mm}^2$  phase conductor (same as Chi
# Conductor mass:  $\text{Cu} = 3 \times 35\text{mm}^2 + 1 \times 25\text{mm}^2 = 130\text{mm}^2$  |  $\text{Al} = 3 \times 50\text{mm}^2 + 1 \times$ 
# Non-material cost differs from China due to European labor rates
# =====

{
  "region": "Europe",
  "dominant material": "cu",
  "product": "Copper Service Entrance Cable,  $35\text{mm}^2$ , 3C+G, 0.6/1kV, 1.5m Drip Loop",
  "copper_kg": 1.75,      #  $130\text{mm}^2 \times 1.5\text{m} \times 8.960\text{ g/cm}^3 = 1,747\text{g} \rightarrow 1.75\text{ kg}$ 
  "aluminum_kg": 0.00,
  "total_mass_kg": 2.80,  #  $1.75\text{kg Cu} \div 0.62$ 
  "non_material_cost_per_unit": 80.00, # USD: European labor + XLPE insulation + fittings
  "attribute_2_value": 1,   # material_continuity: copper history  $\rightarrow 1$ 
  "attribute_3_value": 35.0, # conductor_gauge_mm2: IEC standard  $35\text{mm}^2$ 
  "attribute_4_value": 1.50, # cable_length_m
  "attribute_5_value": 100,  # service_amperage_A: European standard residential 100A
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "Aluminum Service Entrance Cable,  $50\text{mm}^2$ , 3C+G, 0.6/1kV, 1.5m Drip Loop",
  "copper_kg": 0.00,
  "aluminum_kg": 0.75,      #  $185\text{mm}^2 \times 1.5\text{m} \times 2.700\text{ g/cm}^3 = 749\text{g} \rightarrow 0.75\text{ kg}$ 
  "total_mass_kg": 1.40,    #  $0.75\text{kg Al} \div 0.52$ 
  "non_material_cost_per_unit": 80.00, # Same European labor premium
  "attribute_2_value": 0,    # material_continuity: aluminum history  $\rightarrow 0$ 
  "attribute_3_value": 50.0, # conductor_gauge_mm2: IEC standard  $50\text{mm}^2$  (
  "attribute_4_value": 1.50, # cable_length_m
  "attribute_5_value": 100,  # service_amperage_A
},

```

